SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

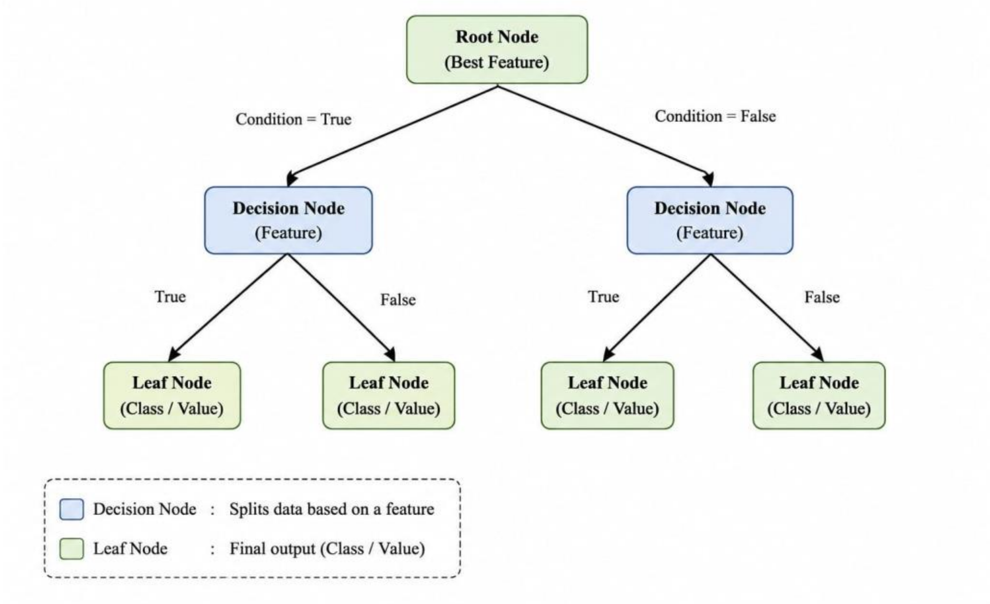
SCENARIO-BASED QUESTIONS

Code of Conduct:

I __Chalamala Sai Harshitha (192424346)__ *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

1. **With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models.**



A **Decision Tree** is a supervised machine learning algorithm that makes predictions by splitting data into branches based on feature values until a final decision or class label is reached.

Feature Selection using Entropy and Information Gain

Entropy:

Entropy measures the impurity or randomness of a dataset. A high entropy value indicates that the data contains mixed classes, whereas an entropy value of zero indicates that all samples belong to the same class (pure node).

Information Gain:

Information Gain measures the reduction in entropy after splitting the dataset using a feature. The feature with the highest information gain is selected for splitting because it produces the purest child nodes and improves the decision-making process.

Tree Construction

Example:

Consider the following dataset:

Weather	Play Tennis
Sunny	No
Rainy	Yes
Overcast	Yes
Sunny	No
Rainy	Yes

The algorithm calculates the entropy for each feature and selects the feature with the highest information gain as the root node.

Example:

```
          Weather
         /   |   \
    Sunny  Rainy  Overcast
      No    Yes     Yes
```

The process continues recursively until all leaf nodes contain pure classes or a stopping condition is satisfied.

Pruning

Pruning is the process of removing unnecessary branches from a decision tree to improve its performance and reduce complexity.

Pre-pruning: Stops the tree from growing beyond a specified depth or minimum number of samples.

Post-pruning: Builds the complete tree first and then removes branches that do not improve accuracy.

Controlling Overfitting

Overfitting occurs when the decision tree becomes too complex and memorizes the training data instead of learning general patterns.

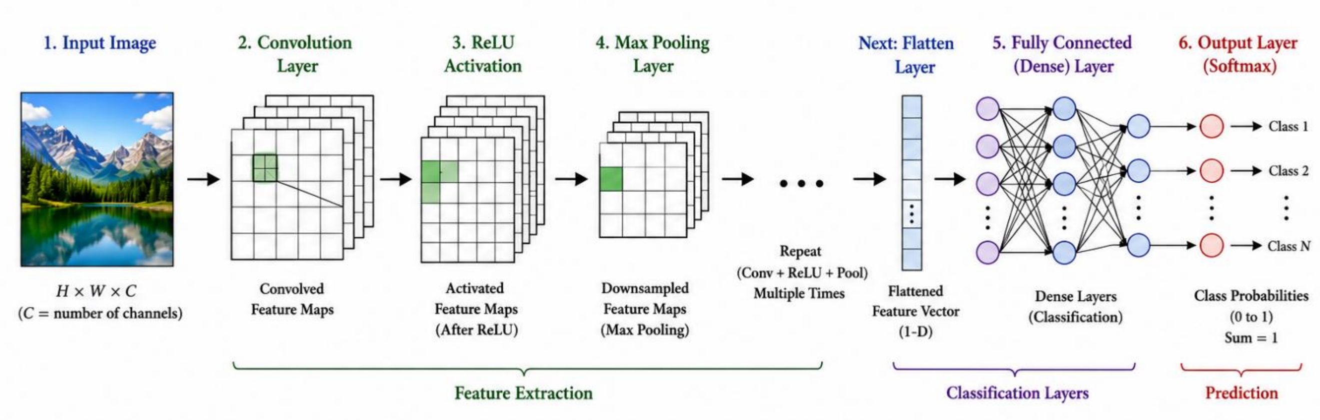
Overfitting can be controlled by:

- Limiting the maximum depth of the tree.
- Setting the minimum number of samples required to split a node.
- Applying pre-pruning or post-pruning techniques.
- Using cross-validation to evaluate the model.

Conclusion

A Decision Tree Learning System selects the best features using entropy and information gain, constructs the tree by recursively splitting the dataset, and uses pruning techniques to reduce overfitting. This improves the model's accuracy, interpretability, and ability to generalize to unseen data.

2. **In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance**.



A **Convolutional Neural Network (CNN)** is a deep learning algorithm primarily used for image processing and computer vision tasks such as image classification, object detection, and image recognition. CNNs automatically learn important features from images through multiple layers, reducing the need for manual feature extraction.

Layers of a Convolutional Neural Network

1) Convolutional Layer

The convolutional layer is the first and most important layer of a CNN. It applies filters (kernels) to the input image to detect features such as edges, corners, textures, and shapes. Each filter extracts a different feature, producing a feature map.

2) Pooling Layer

The pooling layer reduces the size of the feature maps while retaining the most important information. This reduces computational complexity and helps prevent overfitting. The most commonly used pooling operation is **Max Pooling**, which selects the maximum value from each region.

3) Fully Connected Layer

The fully connected layer receives the extracted features from the previous layers and performs classification. Every neuron is connected to every neuron in the previous layer, and the final output layer predicts the class of the input image.

4) Feature Map Generation

A feature map is generated when a convolution filter slides across the input image and performs convolution operations. Different filters extract different visual features, allowing the CNN to learn complex image patterns. As the network becomes deeper, the extracted features become more meaningful and representative of the objects in the image.

5) Backpropagation in CNN

Backpropagation is used to update the weights of the CNN during training.

The process involves:

- Performing a forward pass to generate predictions.
- Calculating the error by comparing predicted and actual outputs.
- Propagating the error backward through the network.
- Updating the filter weights using gradient descent to minimize the loss.

This process is repeated until the model achieves satisfactory accuracy.

Example: Image Classification

Consider a CNN trained to classify images of **cats** and **dogs**.

- The input image is passed through convolutional layers to extract features.
- Pooling layers reduce the feature map dimensions.
- Fully connected layers analyze the extracted features.
- The output layer predicts whether the image belongs to the **Cat** or **Dog** class.

Thus, CNN automatically learns distinguishing features and accurately classifies the image.

Factors Affecting Model Performance

The performance of a CNN depends on several factors:

- Quality and size of the training dataset.
- Number and size of convolution filters.
- Choice of activation function.
- Learning rate and optimization algorithm.
- Number of training epochs.
- Pooling strategy.
- Regularization techniques such as dropout.
- Data augmentation to improve generalization.

Conclusion

Convolutional Neural Networks are powerful deep learning models for image classification and computer vision applications. By using convolutional, pooling, and fully connected layers along with backpropagation, CNNs automatically learn meaningful image features and achieve high classification accuracy across a wide range of applications.

3. **In the context of classification problems, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.**

Introduction

In classification problems, a machine learning model predicts whether an input belongs to a particular class. To evaluate the performance of the model, different evaluation metrics are used. These metrics are calculated using a **confusion matrix** and help measure the correctness and reliability of the model. The most commonly used metrics are **Accuracy, Precision, Recall, F1-Score, and ROC Curve**.

Confusion Matrix

A confusion matrix summarizes the prediction results of a classification model.

- **True Positive (TP):** Correctly predicted positive cases.
- **True Negative (TN):** Correctly predicted negative cases.
- **False Positive (FP):** Negative cases incorrectly predicted as positive.
- **False Negative (FN):** Positive cases incorrectly predicted as negative.

Evaluation Metrics

1. Accuracy

Definition:

Accuracy measures the overall percentage of correctly classified instances.

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

2. Precision

Definition:

Precision measures how many predicted positive cases are actually positive.

Formula:

Precision = TP / (TP + FP)

3. Recall (Sensitivity)

Definition:

Recall measures how many actual positive cases are correctly identified.

Formula:

Recall = TP / (TP + FN)

4. F1-Score

Definition:

F1-Score is the harmonic mean of Precision and Recall.

Formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

5. ROC Curve (Receiver Operating Characteristic Curve)

Definition:

The ROC Curve is a graphical representation that compares the **True Positive Rate (Recall)** with the **False Positive Rate (FPR)** at different classification thresholds. A curve closer to the top-left corner indicates better model performance.

Formula for False Positive Rate (FPR):

FPR = FP / (FP + TN)

Example

Given:

- True Positive (TP) = **180**
- True Negative (TN) = **150**
- False Positive (FP) = **20**
- False Negative (FN) = **50**

1. Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

= (180 + 150) / (180 + 150 + 20 + 50) = 0.825

= 82.5%

2. Precision

Precision = TP / (TP + FP)

= 180 / (180 + 20)

= 0.90

= 90%

3. Recall (Sensitivity)

Recall = TP / (TP + FN)

= 180 / (180 + 50) = 0.7826

= 78.26%

4. F1-Score

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.90 × 0.7826) / (0.90 + 0.7826)

= 0.837

= 83.7%

5. False Positive Rate (FPR)

FPR = FP / (FP + TN)

= 20 / (20 + 150) = 0.1176

= 11.76%

Importance in Real-World Applications

Medical Diagnosis

- **Recall** is very important because missing a disease (False Negative) can be dangerous.
- A high recall ensures that most patients with the disease are correctly identified.

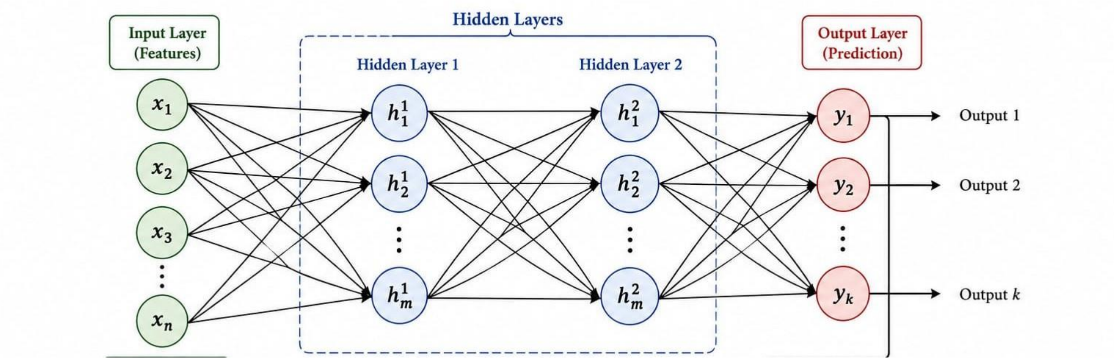
Fraud Detection

- **Precision** is important because wrongly identifying genuine transactions as fraud (False Positive) causes inconvenience to customers.
- A high precision reduces unnecessary fraud alerts.

Conclusion

Evaluation metrics help determine the performance of a classification model. **Accuracy** measures overall correctness, **Precision** measures the correctness of positive predictions, **Recall** measures how many actual positive cases are correctly detected, **F1-Score** balances Precision and Recall, and the **ROC Curve** evaluates the model at different classification thresholds. Together, these metrics provide a complete assessment of a model's performance and help in selecting reliable models for applications such as **medical diagnosis** and **fraud detection**.

4. **With the help of a neat diagram, explain the architecture of an Artificial Neural Network (ANN). Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization.**



An **Artificial Neural Network (ANN)** is a supervised machine learning model inspired by the human brain that learns patterns from data to make predictions or classifications.

ANN Layers:

Input Layer

The input layer is the first layer of the ANN. It receives the input features from the dataset and passes them to the hidden layer without performing any computation. Each neuron in the input layer represents one input feature.

Hidden Layer

The hidden layer performs computations on the input data. It applies weights, biases, and activation functions to learn patterns and relationships within the data. A neural network may contain one or more hidden layers depending on the complexity of the problem.

Output Layer

The output layer produces the final prediction or classification result. The number of neurons in this layer depends on the number of output classes or predicted values.

Weights and Biases

- **Weights** determine the importance of each input feature during learning. They are continuously updated during training to minimize prediction errors.
- **Biases** provide additional flexibility by shifting the activation function, enabling the model to learn more effectively.

Forward Propagation

Forward propagation is the process of passing input data through the neural network to generate the output.

The steps involved are:

- The input features are provided to the input layer.
- The hidden layer computes weighted sums and applies an activation function.
- The processed information is passed to the output layer.
- The output layer produces the final prediction.

Simple Example

Consider a student performance prediction system.

Input Features:

- Study Hours
- Attendance
- Previous Marks

The hidden layer processes these features and learns their relationships. The output layer predicts whether the student is likely to **Pass** or **Fail**.

Controlling Overfitting

Overfitting occurs when the neural network learns the training data too closely and performs poorly on unseen data.

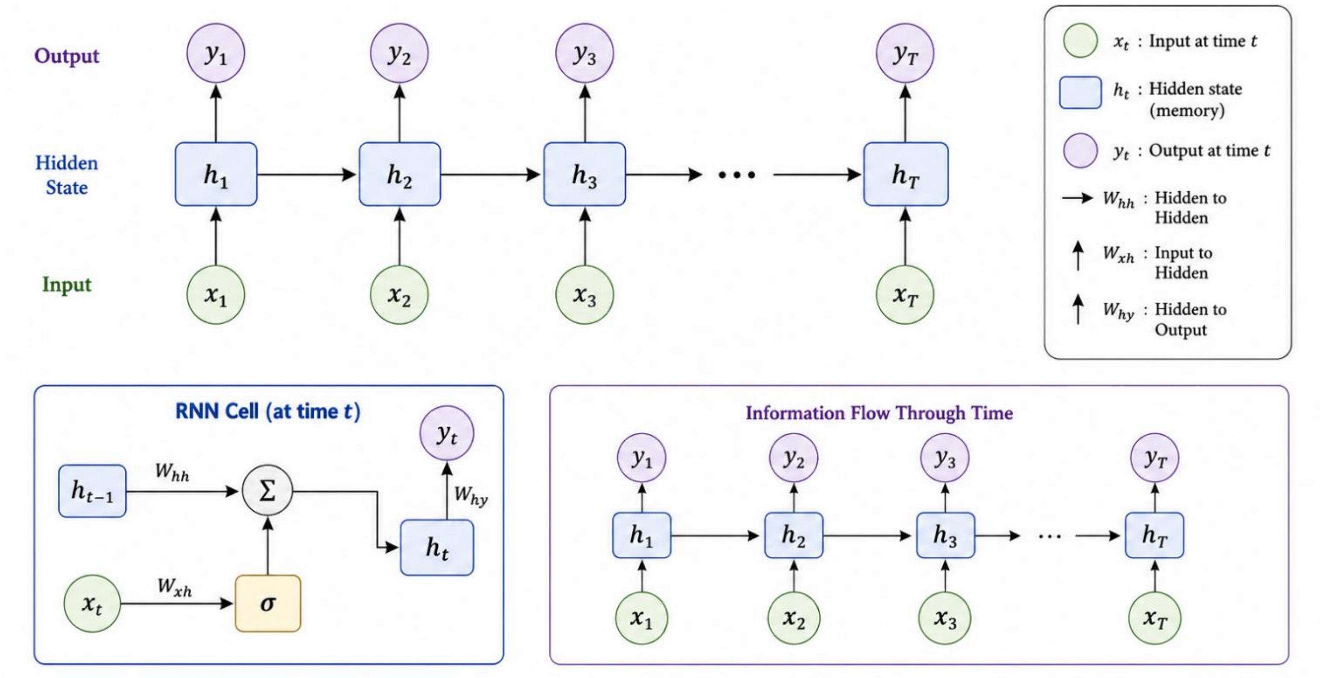
The following techniques help control overfitting:

- **Dropout:** Randomly disables a fraction of neurons during training, preventing the network from relying too heavily on specific neurons.
- **Regularization:** Adds a penalty to the loss function to prevent excessively large weights and improve generalization.
- **Early Stopping:** Stops training when the validation performance no longer improves.
- **Using More Training Data:** Improves the model's ability to generalize.

Conclusion

An Artificial Neural Network consists of input, hidden, and output layers connected through weights and biases. During forward propagation, data flows through these layers to generate predictions. Techniques such as dropout and regularization reduce overfitting and improve the network's accuracy and generalization, making ANNs effective for solving complex machine learning.

5. **In the context of Recurrent Neural Networks (RNNs), explain the architecture and how they handle sequential data. Describe the concept of hidden states and how information is passed through time steps. Illustrate with an example such as text prediction or time-series forecasting, and discuss challenges like vanishing gradients and possible solutions.**



A **Recurrent Neural Network (RNN)** is a type of deep learning model designed to process **sequential data**. Unlike traditional neural networks, RNNs have memory that allows information from previous time steps to influence the current output. They are widely used in applications such as text prediction, speech recognition, language translation, and time-series forecasting.

RNN Architecture

An RNN consists of an input layer, recurrent hidden layer, and output layer. The hidden layer receives both the current input and the previous hidden state, allowing the network to retain information from earlier time steps. This enables the model to capture sequential dependencies in the data.

Hidden States

The **hidden state** acts as the memory of the RNN. At each time step, it stores information from the previous inputs and combines it with the current input. This memory helps the network understand the context of the sequence and make more accurate predictions.

Information Flow Through Time Steps

In an RNN, the input is processed one element at a time. At each time step:

- The current input is received.
- The previous hidden state is combined with the current input.

- A new hidden state is generated.
- The output is produced.
- The updated hidden state is passed to the next time step.

This sequential flow allows the network to learn relationships between earlier and later elements.

Example: Text Prediction

Consider the sentence:

"Machine Learning is ____"

The RNN processes each word one after another while remembering the previous words through hidden states. Based on the learned context, it predicts the next word, such as **"powerful"**, **"important"**, or **"useful"**.

Similarly, in time-series forecasting, an RNN uses previous observations to predict future values.

Challenges in RNN

Vanishing Gradient:

During training, the gradients may become extremely small as they are propagated backward through many time steps. As a result, the network struggles to learn long-term dependencies, reducing prediction accuracy.

Exploding Gradient:

Sometimes, gradients become excessively large during training, causing unstable weight updates and making the learning process difficult.

Solutions

The challenges of RNNs can be addressed using:

- **Long Short-Term Memory (LSTM):** Uses memory cells and gates to retain important information for longer periods.
- **Gated Recurrent Unit (GRU):** A simplified version of LSTM that efficiently captures long-term dependencies with fewer parameters.
- **Gradient Clipping:** Prevents exploding gradients by limiting their magnitude.

Conclusion

Recurrent Neural Networks are powerful deep learning models for processing sequential data. By using hidden states, they retain information from previous time steps and make accurate predictions for tasks such as text prediction and time-series forecasting. Advanced architectures like LSTM and GRU overcome the limitations of traditional RNNs, improving performance on long sequences.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand